AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 1 · STUDENT LESSON

Precision in Algorithms

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

AI-1 Weeks 9–12: trace ordered instructions and repair missing steps.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Translate a short paper calculation into a Python script. Try the recall before opening notes.

Your learning target

I can translate a short paper calculation into a Python script.

I can predict and explain the order of executed statements.

I can keep expected and observed **output** separate when checking a **program**.

Remember before reading

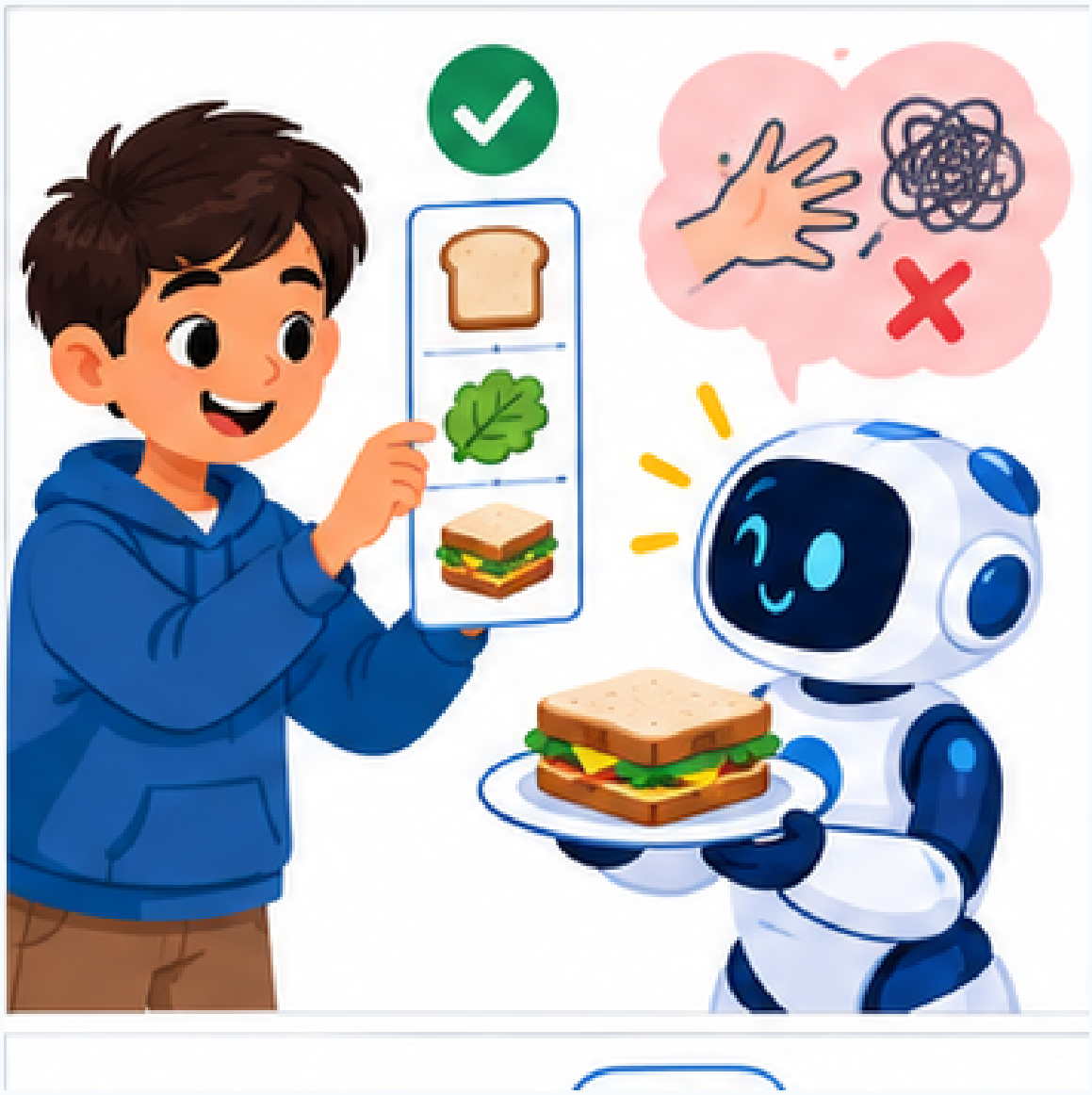
Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Expo and Defense. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 1: Precision in **Algorithm**s

Computers follow stated steps, not intended meaning.



THINKER MISSION

Write two sandwich **algorithm**s; expose hidden **assumption**s.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Four packs contain six counters each. What steps find and show the total?

This week’s end goal

By the end, I can translate an ordered **algorithm** into Python and check state after each line.

STUDENT LESSON · 2 OF 18

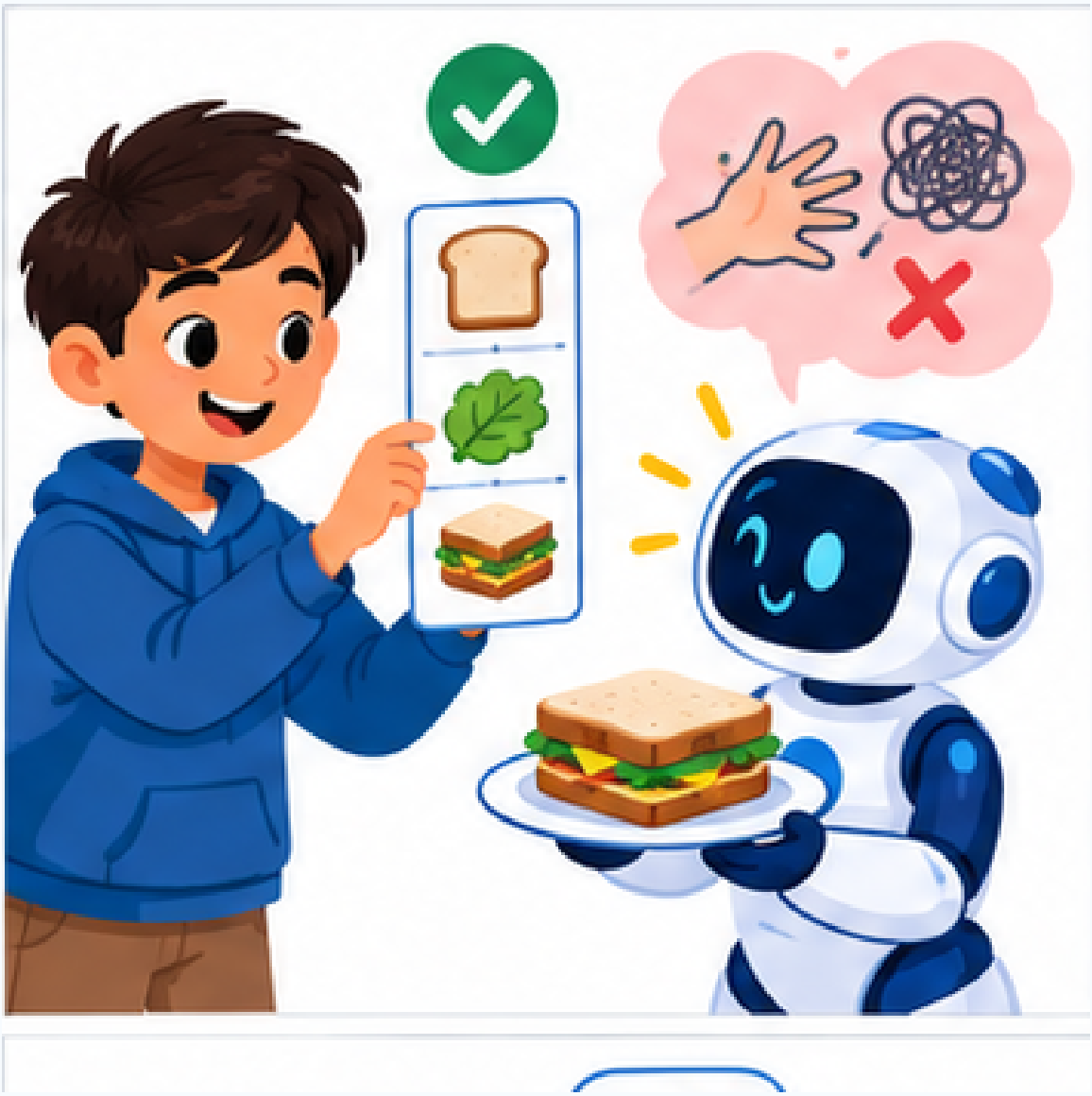
Notice the evidence

Investigate the supplied Precision in Algorithms case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use precision in **algorithm**s to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

In Grade 6 you traced rules on paper. Now you will write a small Python **program** and check what it actually does. Begin with a counter-packing calculation, not the whole activity adviser.

The requirement is to find the total counters in equal packs and display that total. For this first **program**, the supplied quantities are whole, nonnegative numbers; it does not yet validate typed **input**.

Use a teacher-prepared Python 3 editor that can run a saved .py file. Open a new file called counters.py, type the code exactly as shown in the next section, save it, and use the editor's Run command.

Read the **output** area separately from the source editor. No account, AI tool, package installation or personal data is needed. If a device is unavailable, trace on paper and label the result predicted rather than executed.

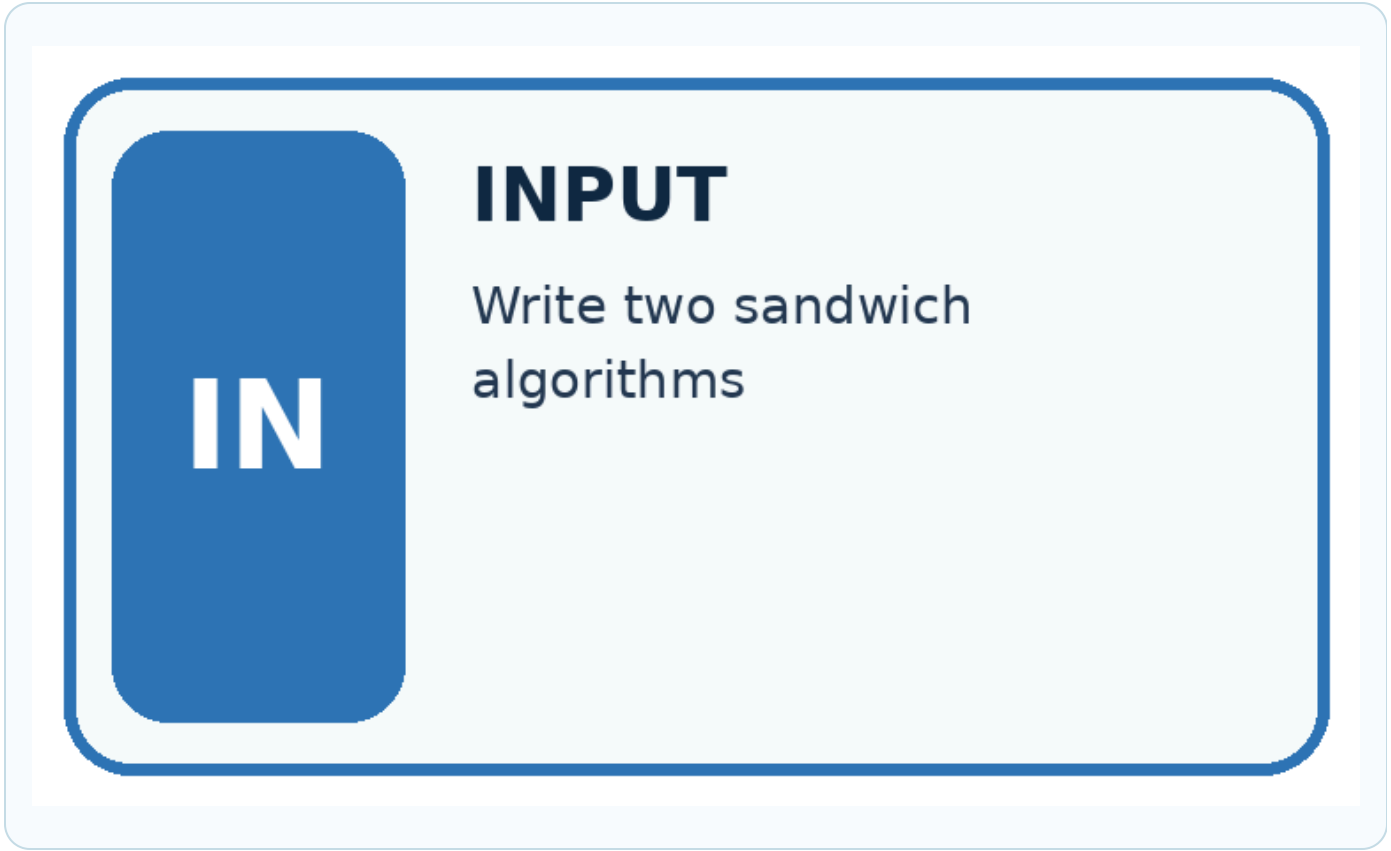
See how it works

> Follow the mechanism for Precision in Algorithms. Connect each labelled input, deciding step and result.

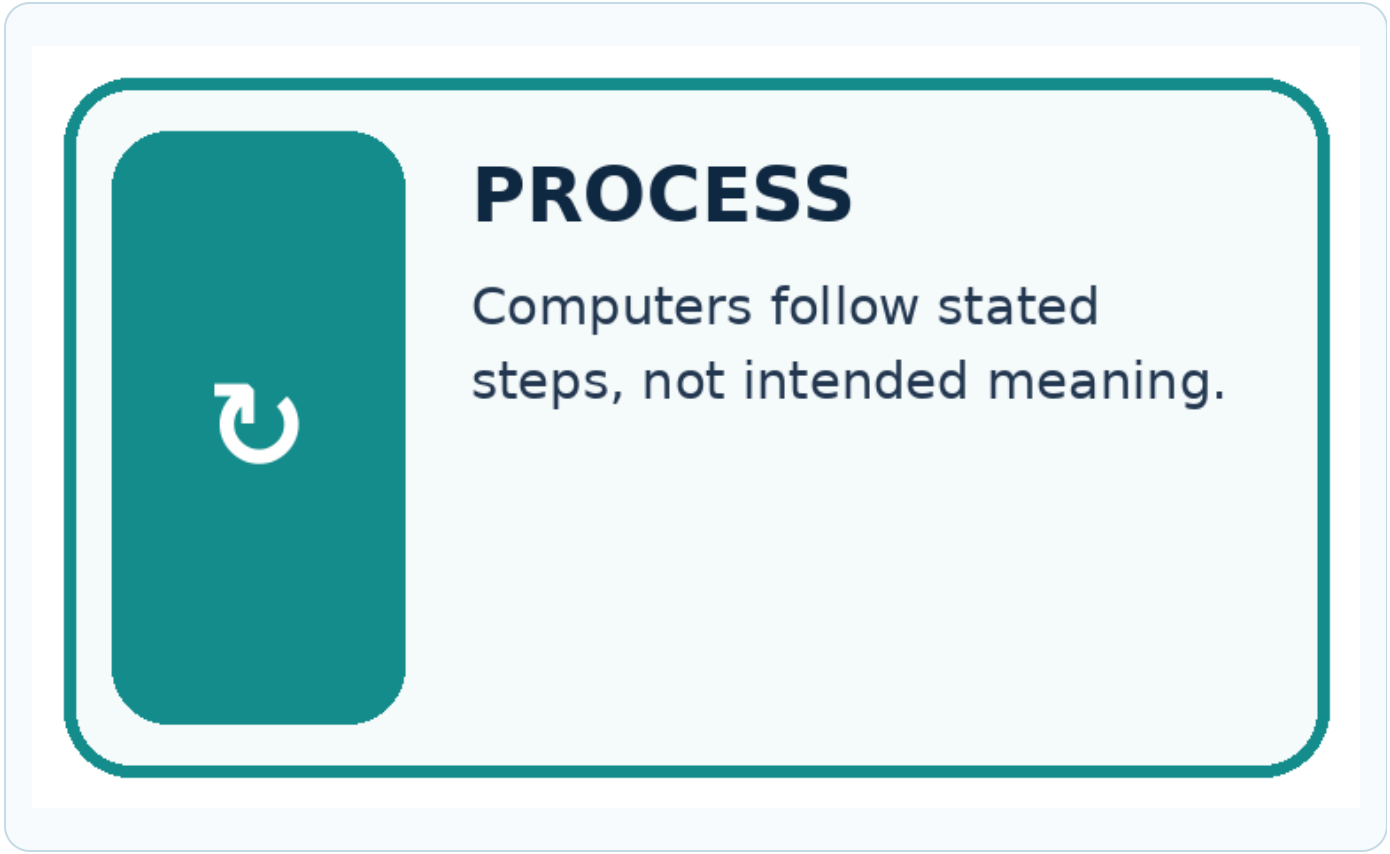
VISUAL MECHANISM

Input → Process → Output

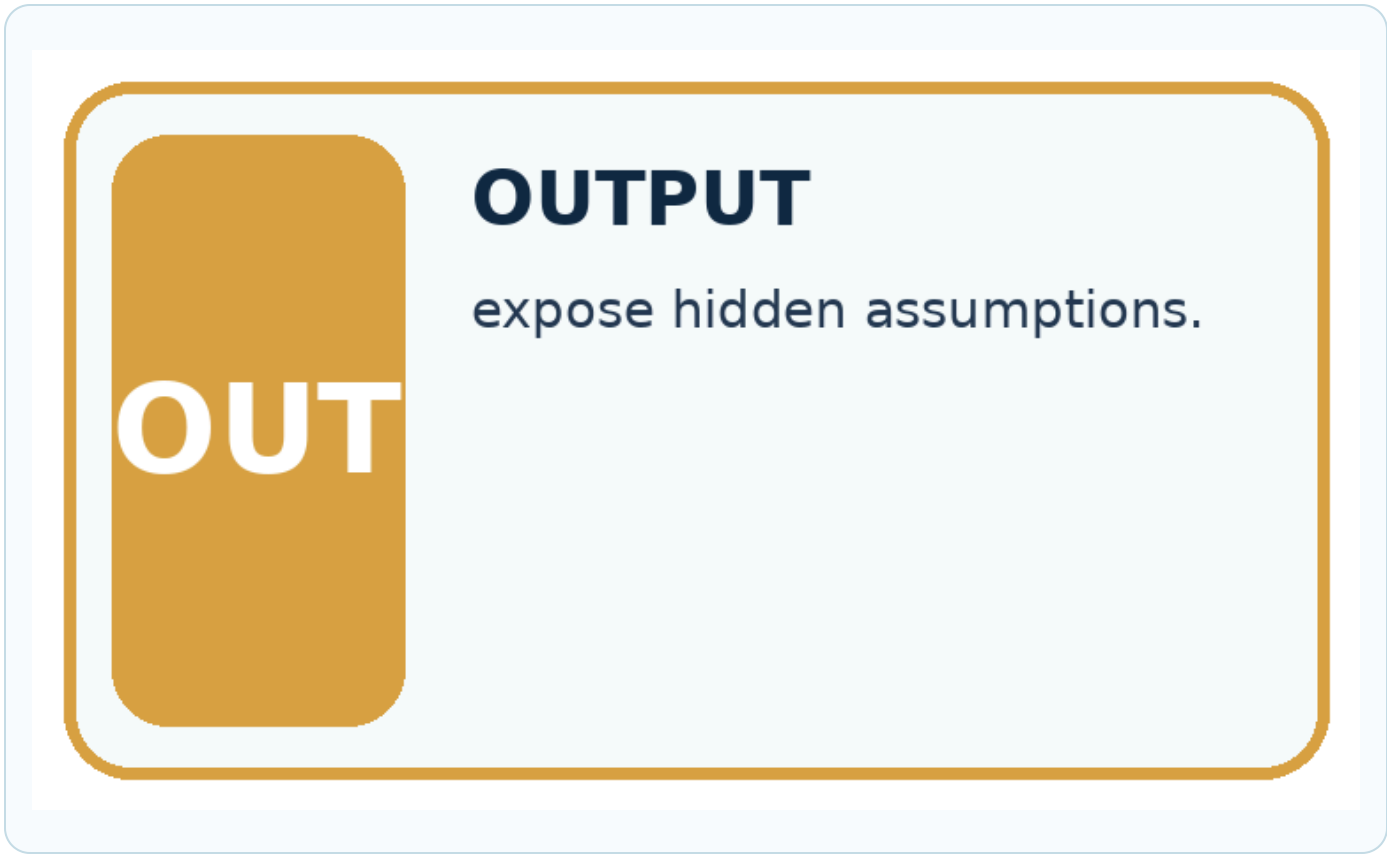
Trace the information. At each stage, ask what changed and how you could test it.



↓


PROCESS
Computers follow stated steps, not intended meaning.

↓


OUT
OUTPUT
expose hidden assumptions.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

A calculation from paper to output

PAPER STEP	PYTHON INSTRUCTION	EFFECT
Record pack count	packs = 4	Name holds 4
Multiply two quantities	total = packs * per_pack	Name holds product
Show the answer	print(total)	Displays the value

Assign per_pack before the multiplication. This diagram summarises the steps; the complete script below includes that assignment.

```
packs = 4
per_pack = 6
total = packs * per_pack
print(total)
```

Read the script from top to bottom. The equals sign assigns the value on its right to the name on its left. The star means multiplication. print(total) displays the value of total; it does not display the letters in the name.

Type ordinary straight characters and keep the parentheses. The Python prompt symbols >>> are not part of this saved script.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

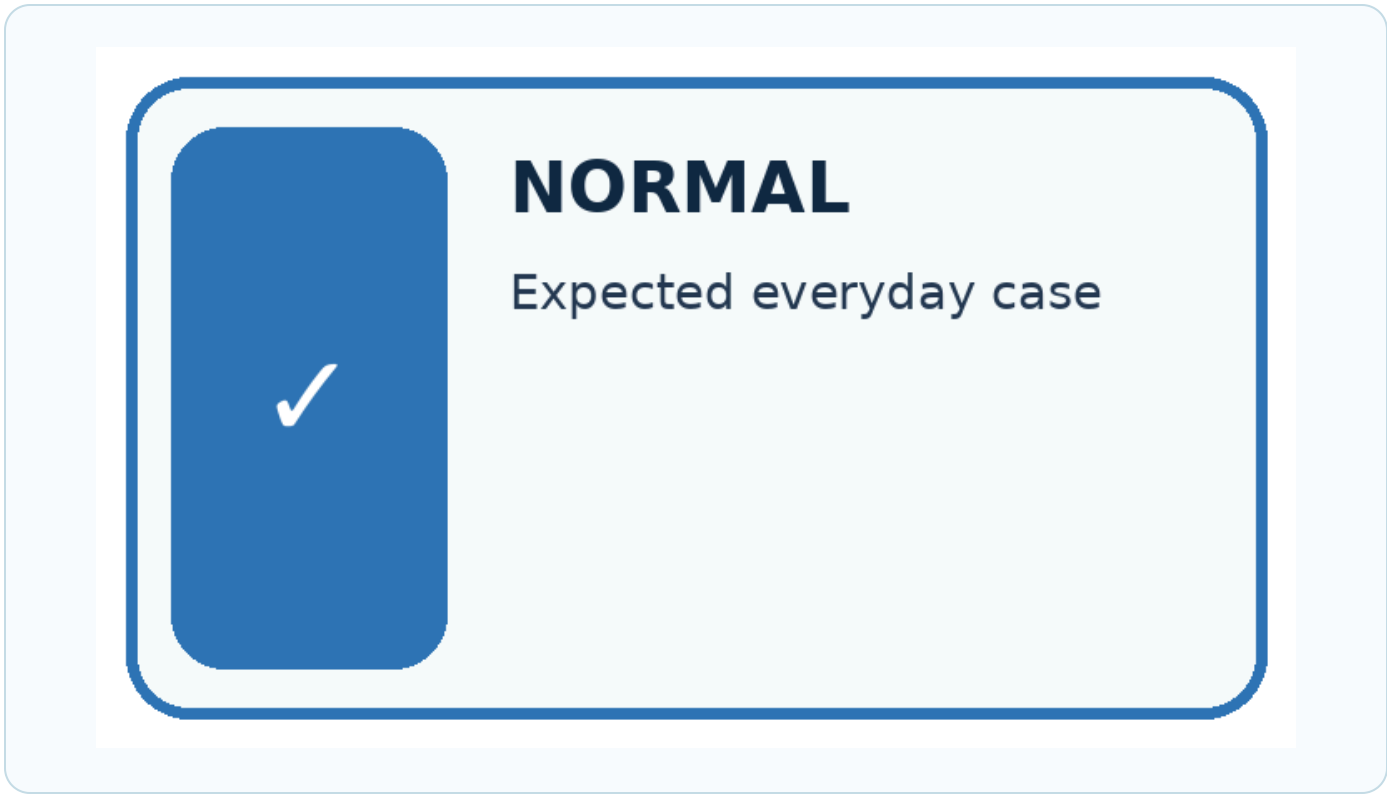
> Compare the supplied cases for Precision in Algorithms. Identify the detail that changes the conclusion.

The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

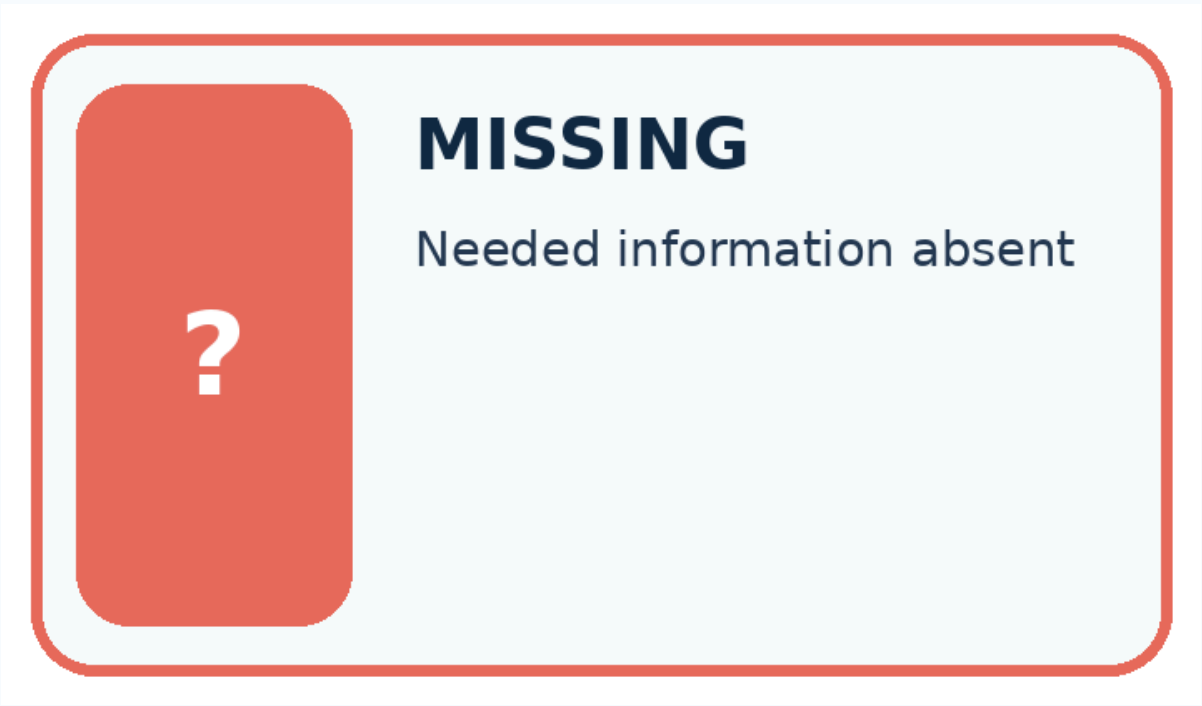
A system is not understood until it survives more than one comfortable example.


NORMAL
Expected everyday case


BOUNDARY
Near a decision edge

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden **assumption**? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Set boxes=0, keeping the two loose books.

Predicted result

Prediction: The printed result is 2.

Why the result follows

Reason: 0×5+2=2; there is no rule saying zero boxes means zero loose books.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

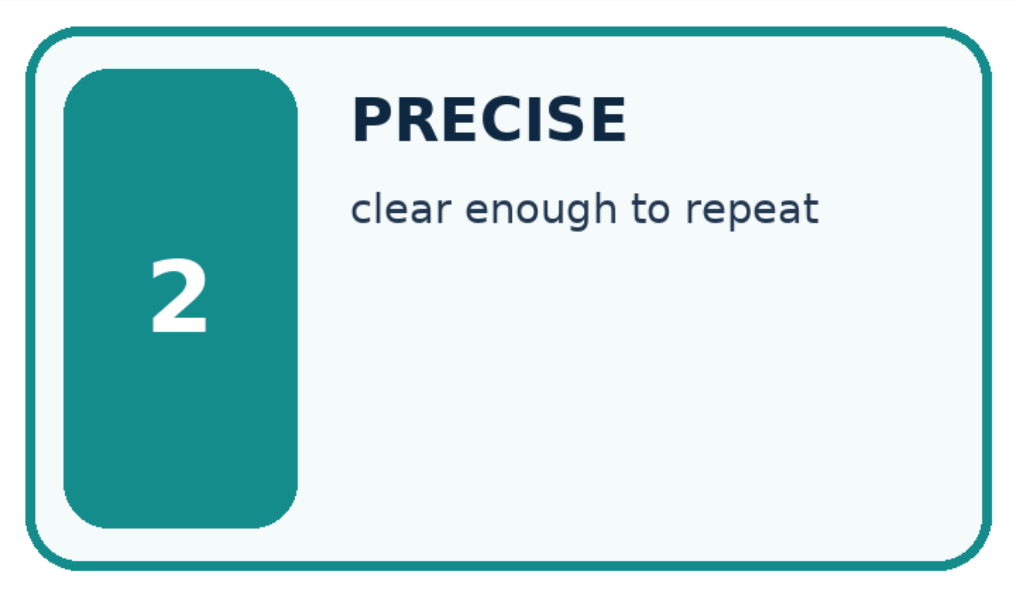
> Use the vocabulary of Precision in Algorithms to explain a concrete decision, rather than only memorising definitions.

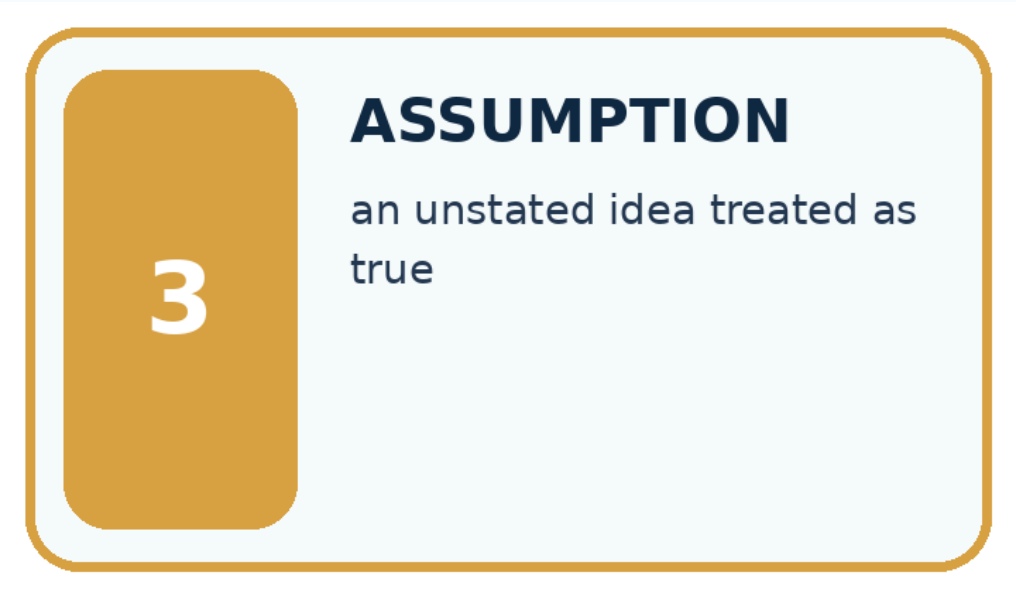
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
ALGORITHM
ordered steps toward a goal


2
PRECISE
clear enough to repeat





Word	My own example	Not this / boundary
algorithm	A specified sequence of steps for performing a task or solving a class of problems.	
precise	Specific enough that a reader can distinguish the intended meaning or reproduce the action.	

Word	My own example	Not this / boundary
assumption	Something treated as true for reasoning that has not been established by the supplied evidence.	
test case	A specified input and conditions with an expected result or measurement rule.	

Words used in the closer explanation

Word	Meaning
Program	Instructions written in a language a computer can execute
Interpreter	Software that executes Python instructions
Assignment	Binding a name to the value of an expression
Output	Information a program displays or returns

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Precision in Algorithms, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 1 • WORKED EVIDENCE

1. Vague: ‘Add the filling.’
2. Question: Which filling? How much? Where?
3. Precise: place 2 cucumber slices on the center of bread A.
4. Test with a new partner: identical results are evidence of precision.

1. Vague: ‘Add the filling.’

2. Question: Which filling? How much? Where?

3. **Precise**: place 2 cucumber slices on the center of bread A.

4. Test with a new partner: identical results are **evidence** of precision.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

After line	Names and values	Displayed output
packs = 4	packs is 4	Nothing yet
per_pack = 6	packs is 4; per_pack is 6	Nothing yet
total = packs * per_pack	total is 24	Nothing yet
print(total)	Values are unchanged	24

Worked explanation · Precision in Algorithms

Write 24 as the expected **output** before running. Then copy the actual **output** into a separate column. A matching result supports this case, not every possible **input**.

A saved file may contain an old edit: save again and rerun the whole script. Do not type the expected number into the **output** record without running.

Why is there no displayed output after the third line? What changes if print(total) is placed first in a fresh run, before any assignment? Explain using which names exist.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Precision in Algorithms

A fictional book-box counter has three boxes with five books each and two loose books.

The facts we will use

Supplied facts

Python instructions: boxes=3; total=boxes*5; total=total+2; print(total).

Supplied facts

Run the instructions in this order with ordinary integer arithmetic.

AI ACADEMY / AI-2/1

Precision in Algorithms

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read input

boxes is assigned 3.



2. Multiply

total becomes 3 x 5=15.



3. Update total

total becomes its current value 15 plus 2: 17.



4. Print

The program prints 17 once.

CHANGE ONE THING

Set boxes=0, keeping the two loose books.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read input	boxes is assigned 3.
2. Multiply	total becomes 3×5=15.
3. Update total	total becomes its current value 15 plus 2: 17.
4. Print	The program prints 17 once.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Assignment replaces a variable's stored value. In total=total+2, Python first reads the current total on the right, calculates 17, then assigns that result to total on the left.

The equal sign is an instruction here, not a claim that 15 and 17 are mathematically equal. Predict **output** before running.

What this example does not establish

Keep the stated limit

These statements count supplied books; they do not validate **input** types or infer stock from measurements.

Keep the stated limit

Read the labelled picture: If print(total) moves before total=total+2, what gets printed for the original **input**?

Keep the stated limit

Change one thing: Set boxes=0, keeping the two loose books. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Precision in Algorithms claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported **assumption**, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A longer instruction is automatically more precise. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Save the file with the **input** values, expected and observed **output**s, and a short explanation. This **program** follows a written multiplication rule.

It has not trained a model, predicted somebody's preferences or checked that real packs contain the stated number. Its conclusion depends on the supplied data. Week 2 examines the values and types behind these names.

A script runs without an error and displays 11 for five packs of six. Is it correct for the stated requirement? Give the required result, the likely wrong operation and one check after repair. State whether your evidence came from a device run or a paper trace.

STUDENT LESSON · 8 OF 18

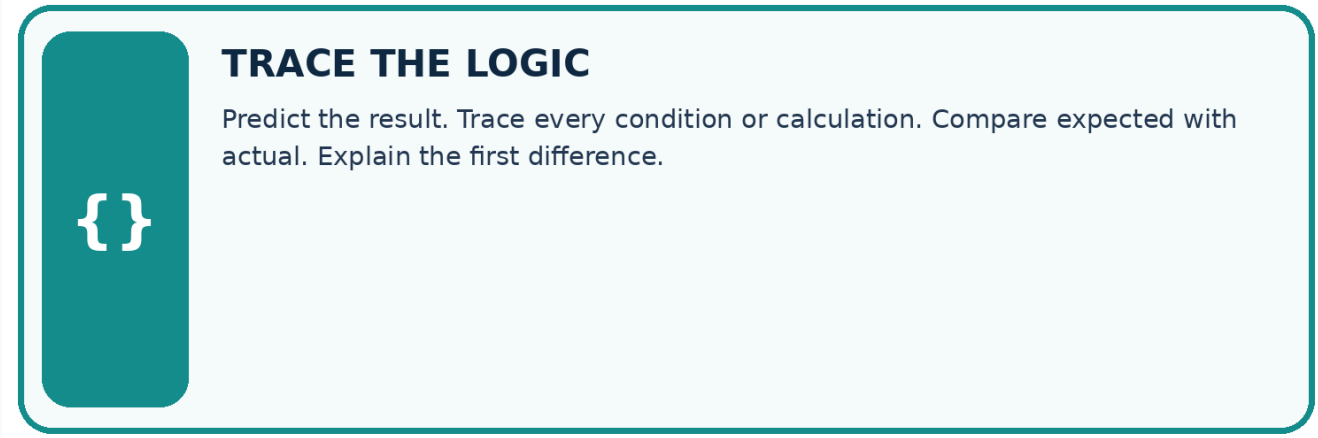
Practise together

Practise Precision in Algorithms with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Change only packs to 5. Predict the result from five groups of six, save, then run the entire file again. The expected **output** is 30. Keep the original four-pack result beside the new one and label the changed **input**.

A comment can explain an instruction: Python ignores text after # outside a string. A comment does not change the calculation.

```
packs = 5  # five equal packs
per_pack = 6
```

```
total = packs * per_pack
print(total)
```

A **program** can run successfully and still solve the wrong problem. Replacing * with + would display 11 for this **input**, but addition does not count five groups of six.

Use the requirement and a paper calculation to check meaning; the absence of an error message is not proof of correctness.

Practise with less help: Predict then observe

Use workbook W2 for the connected example “From a paper **algorithm** to Python”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for From a paper **algorithm** to Python. Proposed explanation: “An equals sign displays the answer”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> If print(total) moves before total=total+2, what gets printed for the original **input**?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

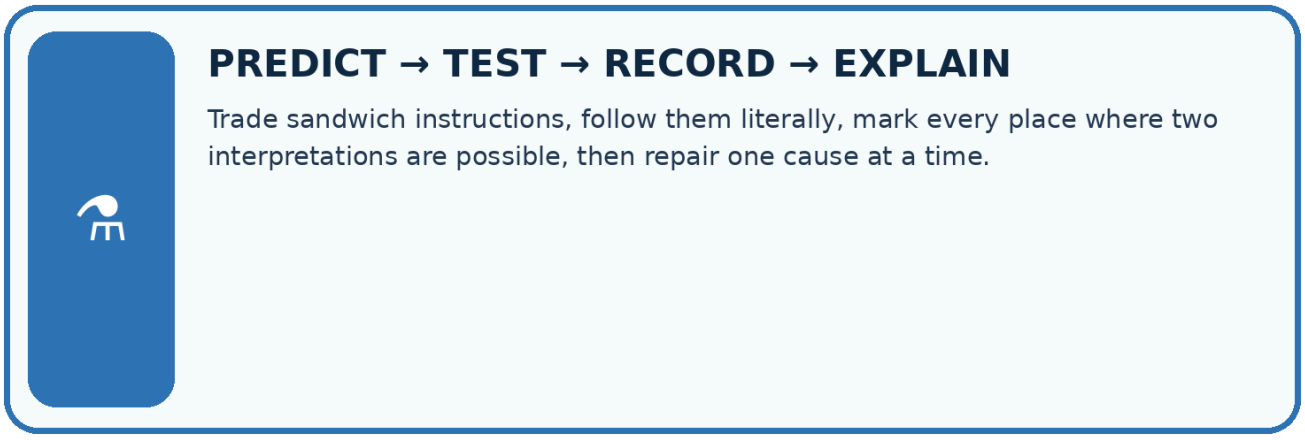
Run the investigation

> Investigate Precision in Algorithms. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Trade sandwich instructions, follow them literally, mark every place where two interpretations are possible, then repair one cause at a time.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

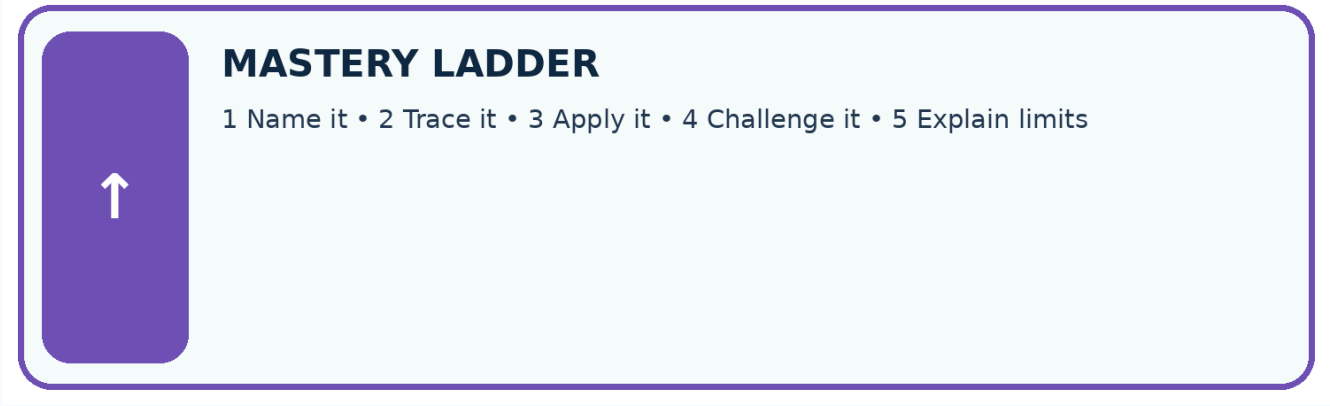
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Precision in Algorithms. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

STUDENT LESSON · 11 OF 18

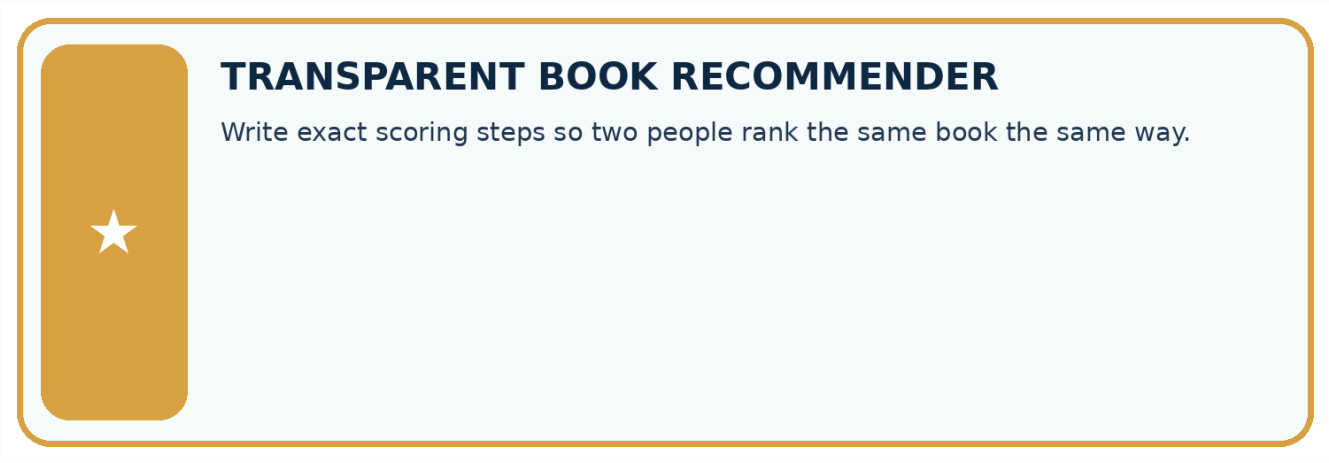
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Write exact scoring steps so two people rank the same book the same way.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Use small predicted traces when translating the Transparent Book Recommender's paper logic into code.

STUDENT LESSON · 12 OF 18

Show what you know

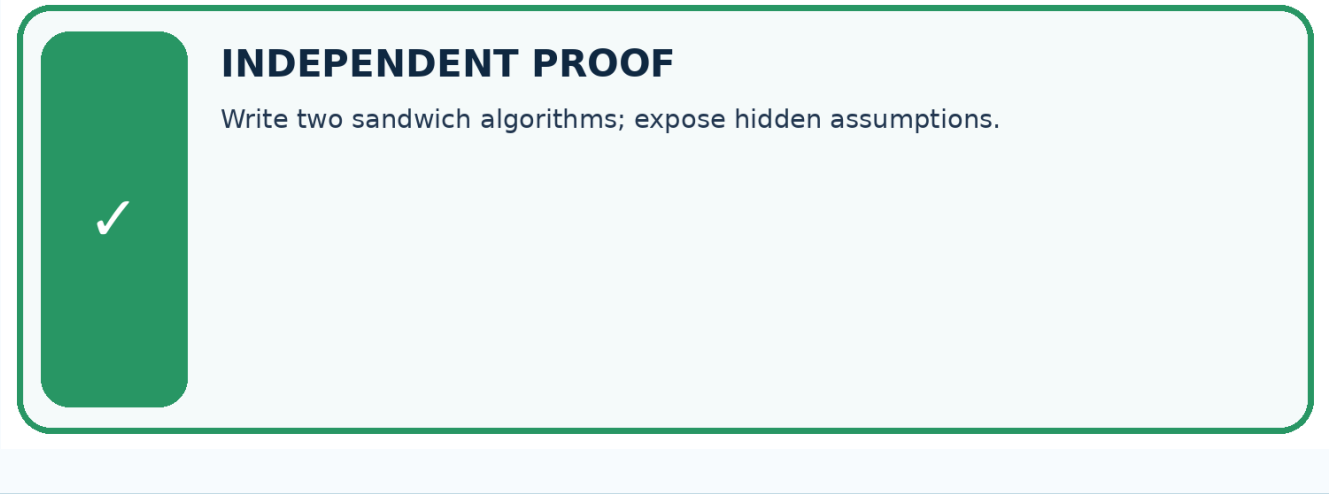
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Write two sandwich **algorithm**s; expose hidden **assumption**s.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Precision in Algorithms case for supported practice. Repeated examples do not become fresh assessments.

Week 1 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A sandwich **algorithm** says add spread, close it, cut it, without stating materials or directions.

The question

Expose **assumption**s and write a **precise** paper simulation.

Worked explanation

Use two paper bread squares, mark a spoon-sized spread area on one, place the second on top and draw a diagonal cut. State materials and order. No real food or sharp tools are needed.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

Trace the supplied case visually

Make hidden instructions explicit

Evidence or stage	Value or result
Materials	Two paper bread squares
Spread	Mark one spoon-sized area
Close	Place second square on top
Cut	Draw a diagonal line

A sandwich **algorithm** says add spread, close it, cut it, without stating materials or directions.

Expose **assumption**s and write a **precise** paper simulation.

Use two paper bread squares, mark a spoon-sized spread area on one, place the second on top and draw a diagonal cut. State materials and order. No real food or sharp tools are needed.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Transfer the procedure independently

A fictional shelf has 3 trays of 7 tiles each. Write a four-line Python script using trays, per_tray and tiles. Predict its **output**. Then change only trays to 0 and predict again. Explain why adding the two quantities is wrong.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 1 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A robot starts at square 0. Replace move a bit and stop nearby with instructions that put it exactly on square 3 using steps of one square.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Write exact scoring steps so two people rank the same book the same way.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Write an **algorithm** for borrowing a classroom book when the book may already be on loan. Include **input**s, a no-copy-available outcome and a check that prevents two borrowers receiving the same copy.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Precision in Algorithms, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional book-box counter has three boxes with five books each and two loose books.

By the end, I can	How I will show it
Translate a short paper calculation into a Python script	Explain the deciding step, show a trace or test, and state one limit.
Predict and explain the order of executed statements	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Keep expected and observed output separate when checking a program	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Precision in Algorithms without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Translate a short paper calculation into a Python script

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or **assumption** and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional book-box counter has three boxes with five books each and two loose books.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The **program** prints 17 once.

1. Read **input**

2. Print

3. Multiply

4. Update total

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Print The **program** prints 17 once. **Assignment** replaces a variable's stored value.

In total=total+2, Python first reads the current total on the right, calculates 17, then assigns that result to total on the left. The equal sign is an instruction here, not a claim that 15 and 17 are mathematically equal.

Predict **output** before running.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Set boxes=0, keeping the two loose books. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The printed result is 2.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The printed result is 2. 0×5+2=2; there is no rule saying zero boxes means zero loose books.

Repair a missing building block

Review Expo and Defense (ai-1/36). Goal: Evaluate a saved candidate and a manual baseline fairly. Short practice: Write one defensible success statement and one claim the **evidence** cannot support..

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

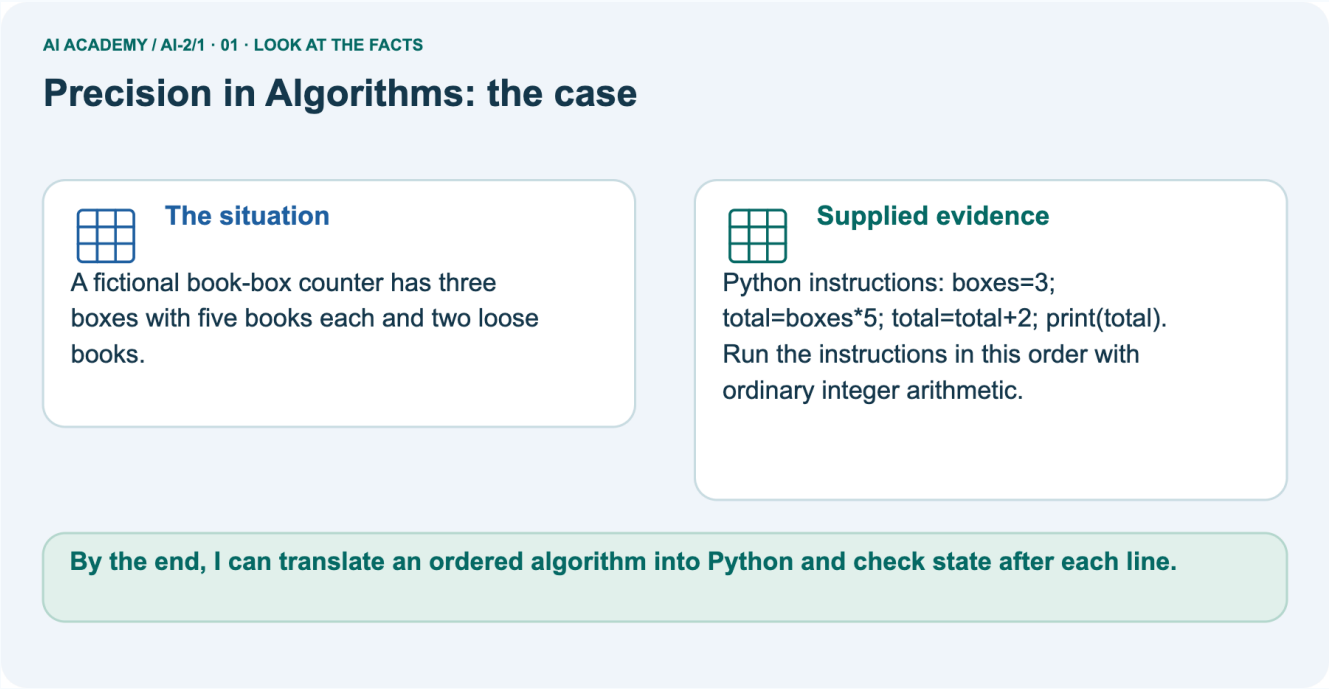
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Precision in Algorithms graphic using its labels, deciding evidence and relevant limitation.

01. Precision in Algorithms: the case



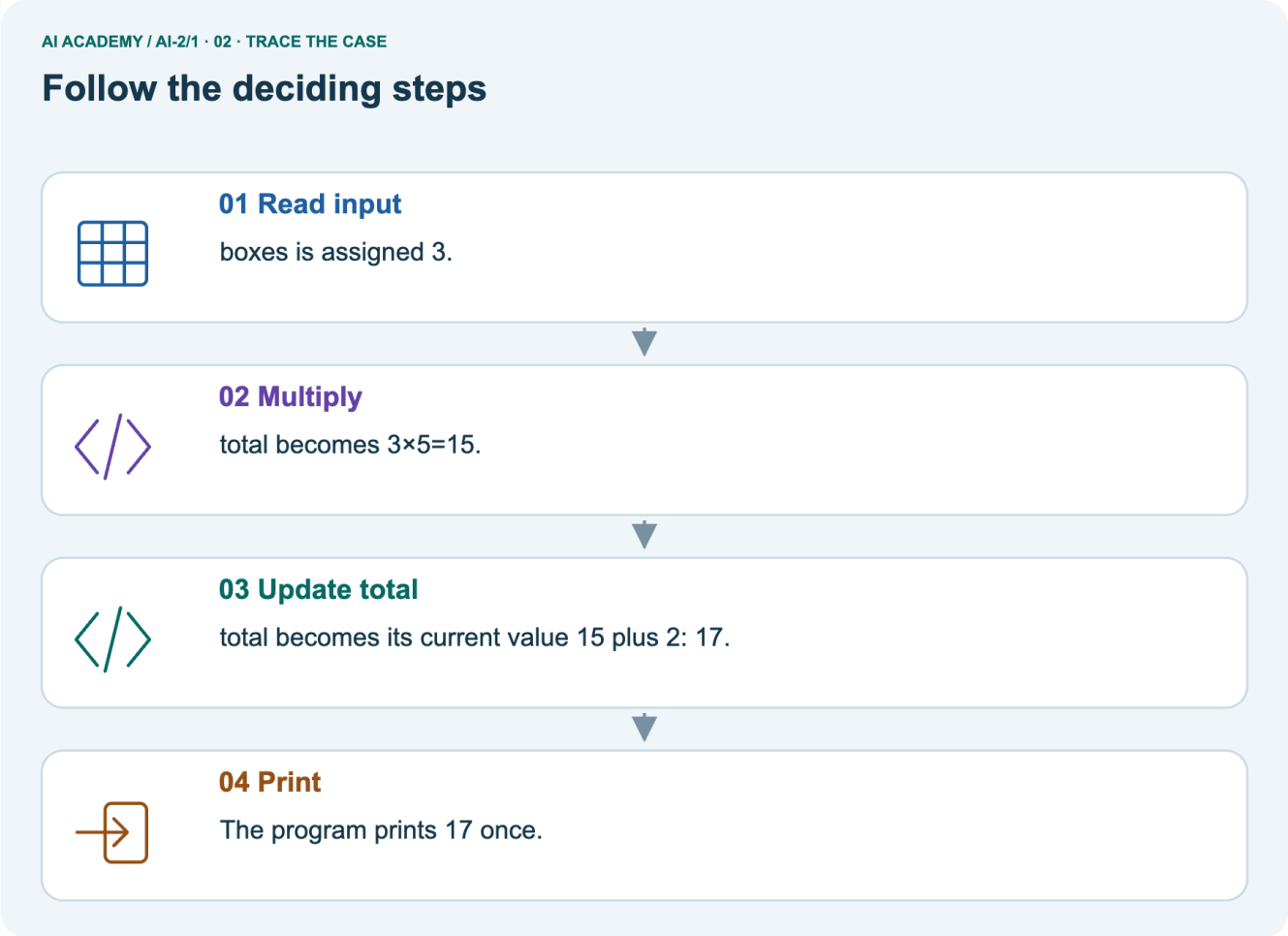
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional book-box counter has three boxes with five books each and two loose books. Supplied facts: Python instructions: boxes=3; total=boxes*5; total=total+2; print(total).

Run the instructions in this order with ordinary integer arithmetic.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read **input**: boxes is assigned 3. → Multiply: total becomes 3×5=15. → Update total: total becomes its current value 15 plus 2: 17. → Print: The **program** prints 17 once.

Point to each numbered stage and explain which recorded **input** it uses.

03. A calculation from paper to output

AI ACADEMY / AI-2/1 · 03 · INSPECT THE LESSON MODEL

A calculation from paper to output

Paper step	Python instruction	Effect
Record pack count	packs = 4	Name holds 4
Multiply two quantities	total = packs * per_pack	Name holds product
Show the answer	print(total)	Displays the value

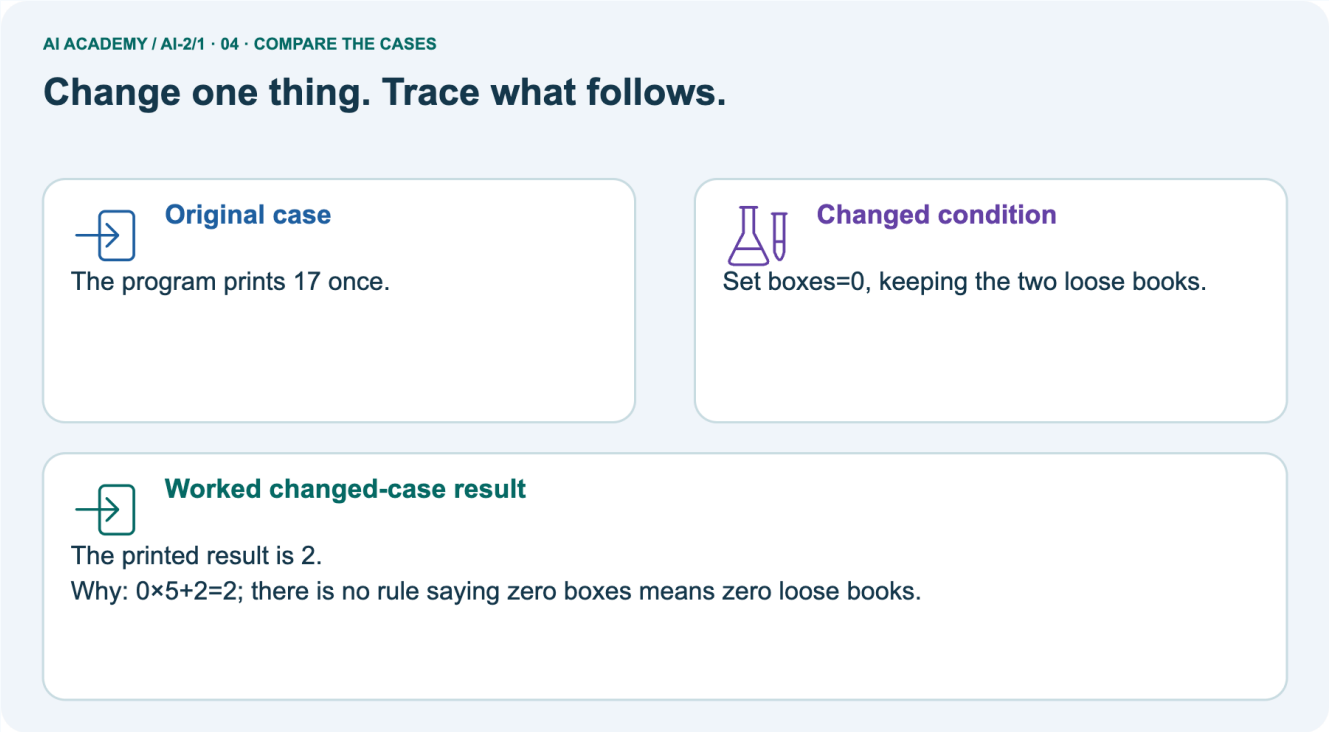
Assign per_pack before the multiplication. This diagram summarises the steps; the complete script below includes that assignment.

Follow the labelled picture with the matching explanation. Open the vector diagram

A calculation from paper to **output**. Record pack count | packs = 4 | Name holds 4. Multiply two quantities | total = packs * per_pack | Name holds product. Show the answer | print(total) | Displays the value

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.

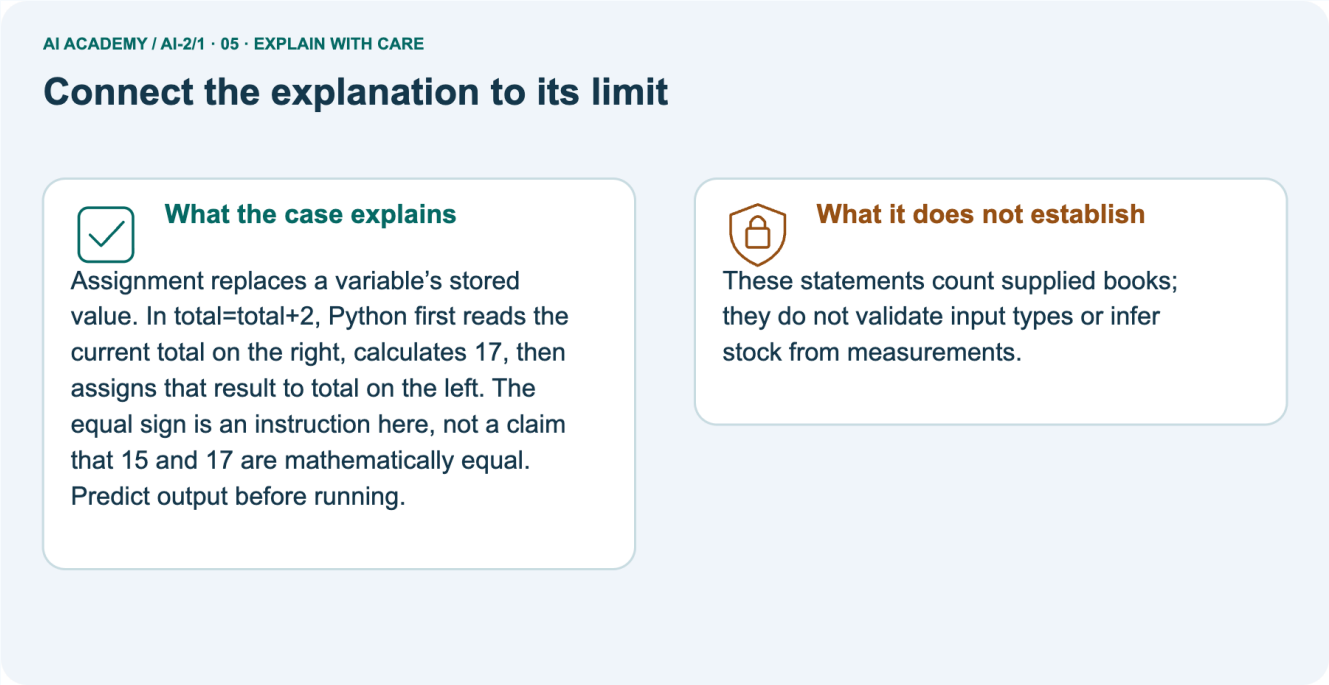


Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The **program** prints 17 once. Changed condition: Set boxes=0, keeping the two loose books. Expected result: The printed result is 2. Reason: 0×5+2=2; there is no rule saying zero boxes means zero loose books.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

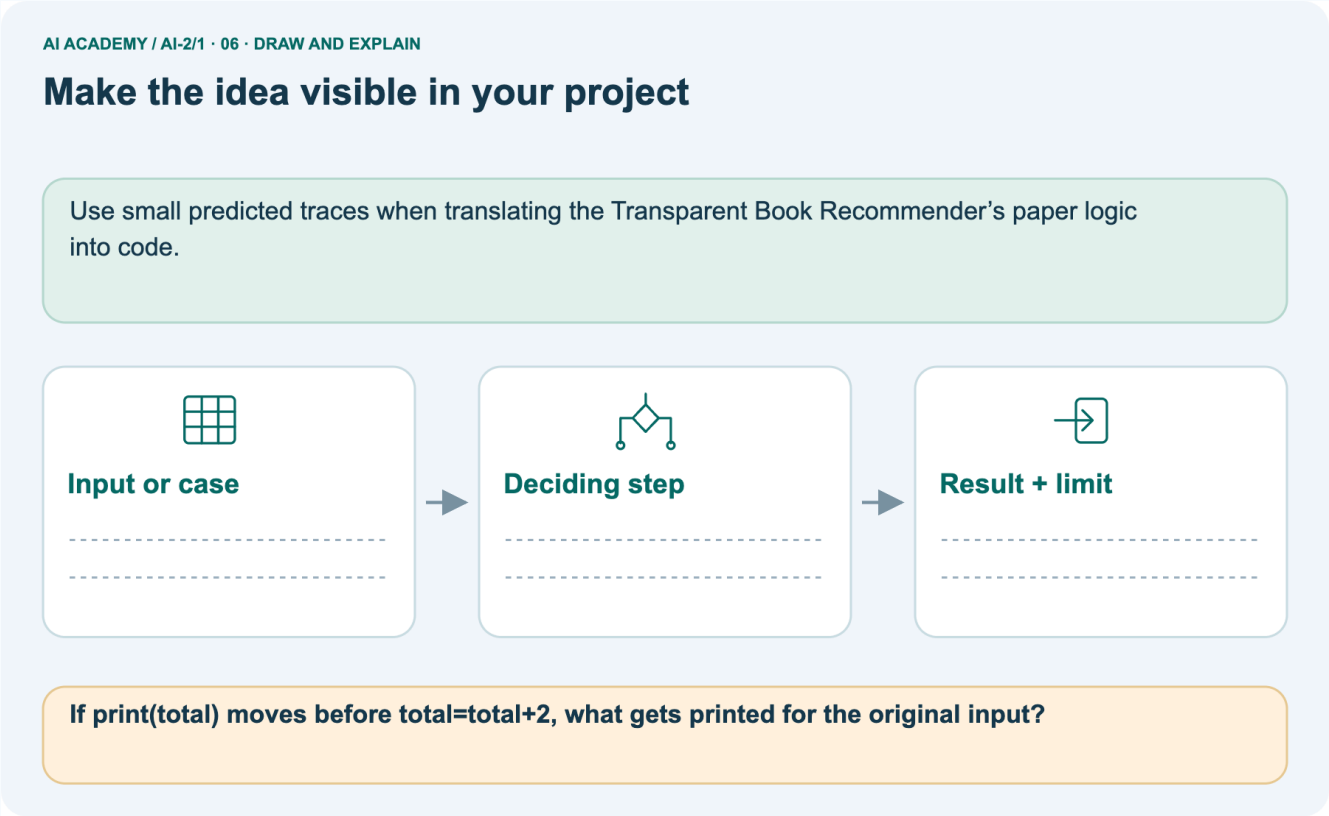
Assignment replaces a variable's stored value. In total=total+2, Python first reads the current total on the right, calculates 17, then assigns that result to total on the left.

The equal sign is an instruction here, not a claim that 15 and 17 are mathematically equal. Predict **output** before running.

Limit: These statements count supplied books; they do not validate **input** types or infer stock from measurements.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use small predicted traces when translating the Transparent Book Recommender's paper logic into code. Draw your **input**, deciding step, result and limit.

Supported question: If print(total) moves before total=total+2, what gets printed for the original **input**?

Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.